

Harsh Singla

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PROFESSIONAL SUMMARY

AI/ML Engineer — Python, TensorFlow, & AI Automation | 1st-year B.Sc. AI/ML student with hands-on experience building ML pipelines, voice AI systems, and agentic automation workflows. Published researcher on n8n automation. Hackathon-proven across 5+ competitions including ISRO's Bharatiya Antariksh initiative. Focused on shipping real, deployable AI solutions.

WORK EXPERIENCE

Corporate Upliftment Bureau (CUB Club) — CGC Landran

Aug 2024 – Present

Member & Event Coordinator

- Coordinated and executed 10+ group discussions, corporate webinars, and industry-connect events, managing logistics for 200+ student participants per session.
- Drove professional development initiatives by inviting industry speakers and facilitating peer-learning workshops aligned with emerging tech trends.

Open to Internship / Freelance Projects — available immediately for part-time or remote ML/AI roles.

TECHNICAL SKILLS

- Languages:** Python, JavaScript (basic), HTML, CSS
- ML / DL:** KNN, K-Means Clustering, Feature Engineering, Data Preprocessing, TensorFlow, Scikit-learn
- Libraries:** NumPy, Pandas, Matplotlib, Seaborn
- AI & Automation:** n8n, Vapi, OmniDimension, Zapier, GPT-4 API, Gemini API, Spotify API
- Tools & Platforms:** VS Code, Git, GitHub, Postman, Kaggle, Streamlit

EDUCATION

Chandigarh Group of Colleges (NAAC A+), Landran

2024 – 2027

Bachelor of Science in Artificial Intelligence & Machine Learning

CGPA: 8.5+ | Relevant Coursework: Machine Learning, Data Structures & Algorithms, Deep Learning, Database Management, Python Programming

PROJECTS

Music Recommendation System [Live Demo](#) [Source Code](#)

Oct 2024 – Nov 2024

- Engineered an end-to-end recommendation pipeline by extracting and preprocessing 10,000+ track audio features via the Spotify API, producing a clean, model-ready dataset.
- Implemented KNN-based collaborative filtering with cosine similarity metrics, delivering personalized track recommendations and improving relevance accuracy by ~18% over a random baseline.
- Applied feature scaling and dimensionality-reduction techniques to reduce noise and improve cluster separability across 8 audio dimensions.

Technologies: Python, Spotify API, KNN, NumPy, Pandas, Scikit-learn, Matplotlib

Lyra – AI Voice Assistant [Demo](#) [Source Code](#)

Dec 2024 – Jan 2025

- Built a fully functional desktop voice assistant using Python, integrating speech-to-text recognition, NLP processing, and multi-step automation workflows.
- Implemented 15+ voice-triggered automation tasks including web search, file management, app control, and calendar queries with <1.5s average response latency.
- Integrated GPT-4 and Gemini APIs to handle open-domain conversational queries, reducing fallback responses by ~30% compared to rule-based approaches.

Technologies: Python, SpeechRecognition, GPT-4 API, Gemini API, Automation Libraries

Moving Wheel – AI-Powered Startup Concept

2025 – Present

- Conceptualized an early-stage mobility-sector startup leveraging AI-driven automation, producing a detailed product specification and technical architecture document.
- Designed n8n-based agentic automation workflows and LLM decision-logic layers to enable intelligent operational routing in the system design.

Technologies: n8n, LLM APIs, System Design, Product Strategy

RESEARCH & PUBLICATIONS

Published research paper analyzing n8n as an open-source agentic automation platform — covering workflow orchestration, multi-step API chaining, conditional logic, and LLM integration for scalable enterprise automation.

ALGORITHMIC COMPETITIONS / ACHIEVEMENTS

-  2nd Place — Delhi University Hackathon (FinTech Track), competing against 100+ teams with a working AI-powered prototype.
- Participant — SAP HackFest | Adobe India Hackathon | Bharatiya Antariksh Hackathon (ISRO Initiative) | Code for Bharat Hackathon
- Certifications: AI Aware, AI Appreciate (Anthropic/Google), freeCodeCamp Full Stack, Algorithm Fundamentals (DSA).
- Solved 100+ Python and algorithmic challenges across online platforms, strengthening core problem-solving fundamentals.